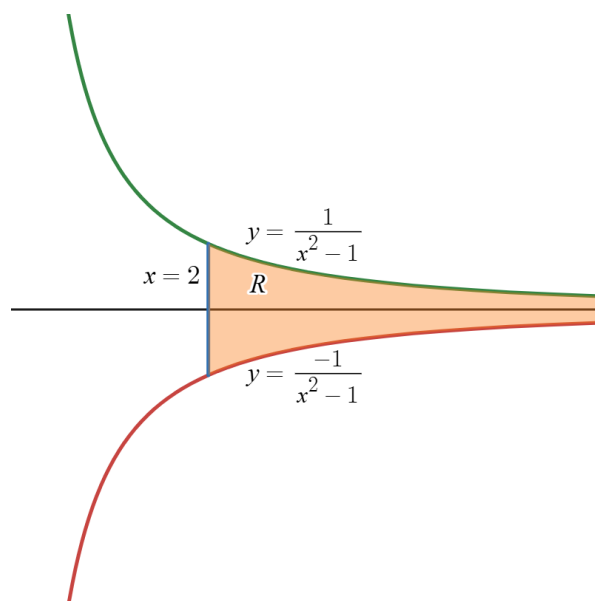


1. (Fall 2022, #3) Consider the infinite planar region R bounded by the vertical line $x = 2$, and the two curves $y = 1/(x^2 - 1)$ and $y = -1/(x^2 - 1)$, as shown in the graph.



- (a) Write down an improper integral that represents the area R , and decide whether it is convergent or divergent;

$$\int_2^{\infty} \left(\frac{1}{x^2-1} - \frac{-1}{x^2-1} \right) dx = \int_2^{\infty} \frac{2}{x^2-1} dx = \int_2^{\infty} \frac{x^2}{x^2-1} \cdot \frac{2}{x^2} dx$$

$$= \int_2^{\infty} \frac{2}{x^2-1} dx \quad \text{convergent}$$

$$= \int_2^{\infty} \frac{1}{1-\frac{1}{x^2}} \cdot \frac{2}{x^2} dx$$

$$\leq \int_2^{\infty} \frac{1}{1-\frac{1}{2^2}} \cdot \frac{2}{x^2} dx$$

$$= \frac{4}{3} \int_2^{\infty} \frac{2}{x^2} dx$$

- (b) Calculate the area of R .

$$\int_2^{\infty} \frac{2}{x^2-1} dx$$

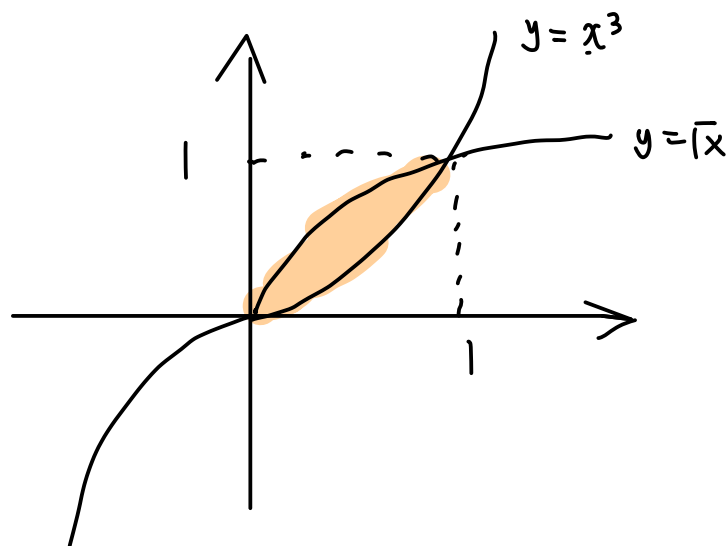
$$= \int_2^{\infty} \frac{2}{(x+1)(x-1)} dx$$

$$= \int_2^{\infty} \left(\frac{1}{x-1} - \frac{1}{x+1} \right) dx$$

$$= \lim_{t \rightarrow \infty} \int_2^t \left(\frac{1}{x-1} - \frac{1}{x+1} \right) dx = \lim_{t \rightarrow \infty} \left(\ln(x-1) - \ln(x+1) \right) \Big|_2^t$$

$$\begin{aligned} &= \lim_{t \rightarrow \infty} \underbrace{\ln\left(\frac{t-1}{t+1}\right)}_{=0} - \ln\left(\frac{2-1}{2+1}\right) \\ &= -\ln\left(\frac{1}{3}\right) \\ &= \ln 3. \end{aligned}$$

2. (Fall 2013, #4a) Let $\mathfrak{R}$ be the region of the plane bounded by the curves $y = x^3$ and $y = \sqrt{x}$. Sketch the region and find its area.



$$\text{Let } \sqrt{x} = x^3 \Rightarrow x=1.$$

$$\mathcal{R} = \int_0^1 (\sqrt{x} - x^3) dx$$

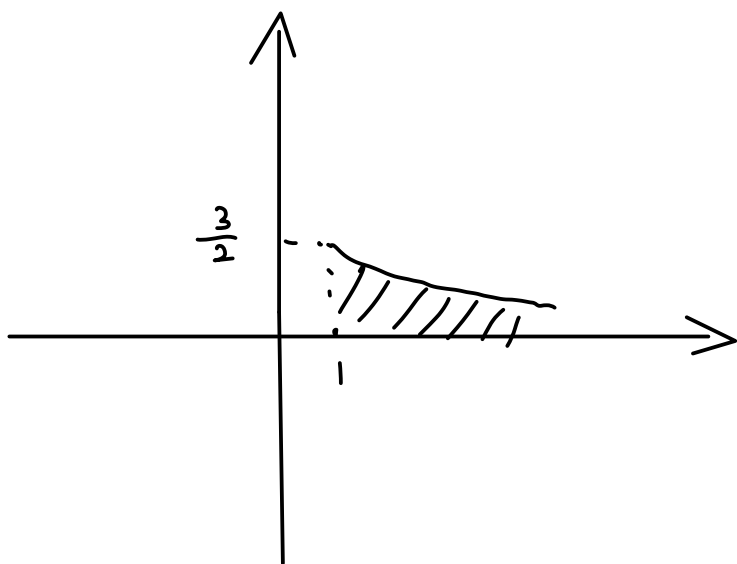
$$= \frac{2}{3} x^{\frac{3}{2}} - \frac{1}{4} x^4 \Big|_0^1$$

$$= \frac{2}{3} - \frac{1}{4} = \frac{5}{12}.$$

3. (Spring 2013, #3) Let A be the region under the graph of $f(x) = \frac{x^2 + x + 1}{x^2(x^2 + 1)}$, above the x -axis, and to the right of $x = 1$. Determine if A has finite area. If so, find it, and if not, explain why.

$$\begin{aligned} f(x) &= \frac{x^2 + x + 1}{x^2(x^2 + 1)} \\ &= \frac{(x^2 + 1) + x}{x^2(x^2 + 1)} = \frac{x^2 + 1}{x^2(x^2 + 1)} + \frac{x}{x^2(x^2 + 1)} \\ &= \frac{1}{x^2} + \frac{1}{x(x^2 + 1)} \end{aligned}$$

when $x > 1$ $f(x)$ is decreasing. $f(1) = 1 + \frac{1}{1 \cdot 2} = \frac{3}{2}$



$$\int_1^{\infty} \left(\frac{1}{x^2} + \frac{1}{x(x^2 + 1)} \right) dx.$$

$$\leq \int_1^{\infty} \frac{1}{x^2} + \frac{1}{x \cdot x^2} dx < \infty. \text{ Finite area.}$$

$$\int_1^{\infty} \left(\frac{1}{x^2} + \frac{1}{x(x^2 + 1)} \right) dx = \int_1^{\infty} \frac{1}{x^2} + \frac{A}{x} + \frac{Bx + C}{x^2 + 1} dx \quad (*)$$

$$= \int_1^{\infty} \frac{1}{x^2} + \frac{A(x^2 + 1) + (Bx + C)x}{x(x^2 + 1)} dx$$

$$= \int_1^{\infty} \frac{1}{x^2} + \frac{(A+B)x^2 + Cx + A}{x(x^2+1)} dx.$$

$$\Rightarrow \begin{cases} A+B=0. \\ C=0. \\ A=1. \end{cases} \Rightarrow \begin{cases} A=1 \\ B=-1 \\ C=0. \end{cases}$$

$$\Rightarrow (*) = \int_1^{\infty} \frac{1}{x^2} + \frac{1}{x} - \frac{x}{x^2+1} dx.$$

$$= \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x^2} + \frac{1}{x} - \frac{x}{x^2+1} dx$$

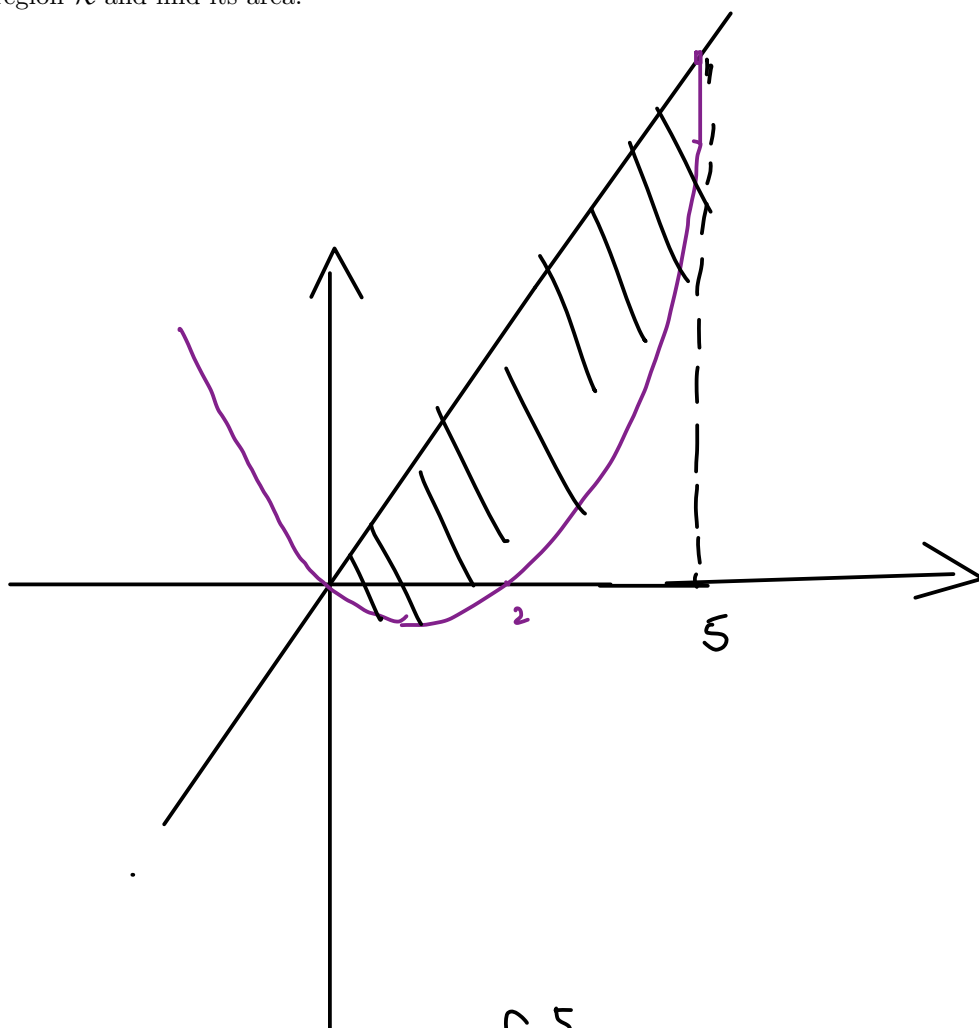
$$= \lim_{t \rightarrow \infty} \left. -\frac{1}{x} + \ln x - \frac{1}{2} \ln(x^2+1) \right|_1^t$$

$$= \lim_{t \rightarrow \infty} \left. -\frac{1}{x} + \ln \frac{x}{\sqrt{x^2+1}} \right|_1^t$$

$$= \lim_{t \rightarrow \infty} \underbrace{-\frac{1}{t}}_{=0} + \underbrace{\ln \frac{t}{\sqrt{t^2+1}}}_{=0} - \left(-1 + \ln \frac{1}{\sqrt{2}} \right) -$$

$$= 1 - \ln \frac{1}{\sqrt{2}} = 1 + \frac{1}{2} \ln 2.$$

4. (Fall 2012, #4i) Consider the region $\mathcal{R}$ between the line $y = 3x$ and the curve $y = x^2 - 2x$. Sketch the region $\mathcal{R}$ and find its area.



$$\begin{cases} y = 3x \\ y = x^2 - 2x \end{cases}$$

$$\Rightarrow x^2 - 2x = 3x$$

$$\Rightarrow x^2 - 5x = 0$$

$$\Rightarrow x(x - 5) = 0$$

$$\Rightarrow x = 0 \text{ or } x = 5$$

$$\mathcal{R} = \int_0^5 3x - (x^2 - 2x) dx$$

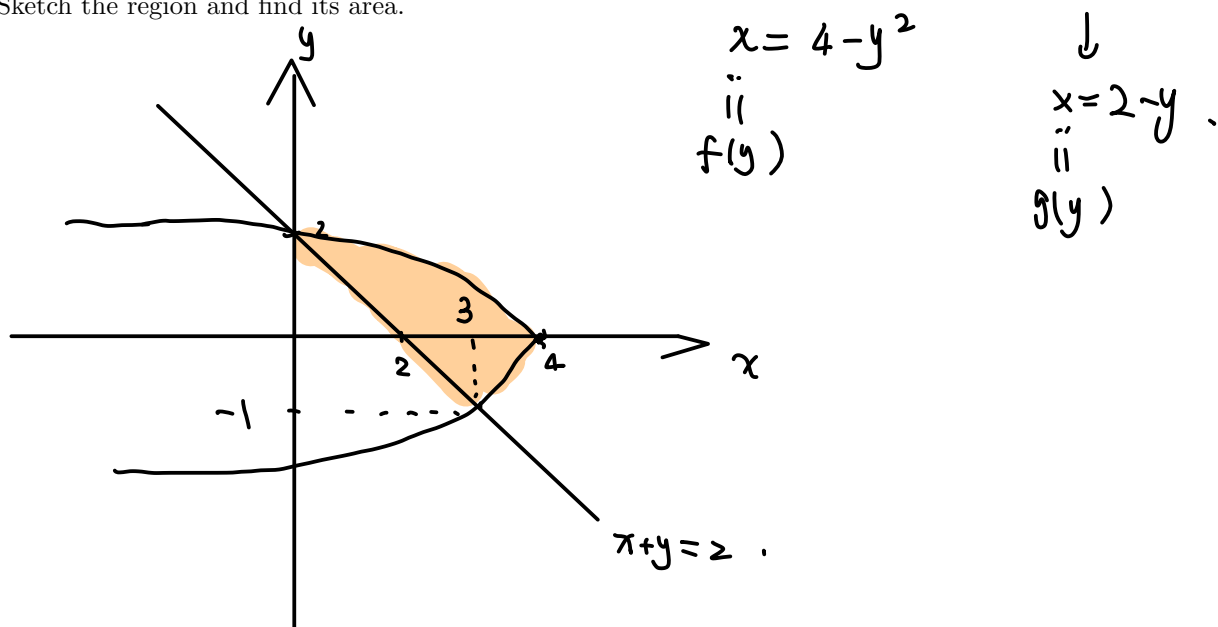
$$= \int_0^5 -x^2 + 5x dx$$

$$= -\frac{x^3}{3} + \frac{5}{2}x^2 \Big|_0^5$$

$$= -\frac{5^3}{3} + \frac{5}{2} \cdot 5^2$$

$$= \frac{125}{6}$$

5. (Spring 2012, #3a) Let $\mathcal{R}$ be the region bounded by the curve $y^2 = 4 - x$ and the line $x + y = 2$. Sketch the region and find its area.



$$\begin{cases} y^2 = 4 - x \\ x + y = 2 \end{cases}$$

The easier way is
to do integration w.r.t y .

$$\Rightarrow 4 - x = (2 - x)^2$$

$$\Rightarrow x^2 - 3x = 0$$

$$\Rightarrow x = 0 \text{ or } x = 3.$$

$$\text{plug in } x = 3 \Rightarrow y = -1$$

$$\text{plug in } x = 0 \Rightarrow y = 2.$$

$$\int_{-1}^2 (4 - y^2) - (2 - y) dy.$$

$$= \int_{-1}^2 -y^2 + y + 2 dy$$

$$= -\frac{y^3}{3} + \frac{y^2}{2} + 2y \Big|_{-1}^2$$

$$= \left(-\frac{8}{3} + \frac{4}{2} + 4\right) - \left(\frac{1}{3} + \frac{1}{2} - 2\right)$$

$$= \frac{9}{2}.$$